

Arc Length and Improper Integrals



Arc length

- 1 Find the arc length of $y = \frac{2}{3}x^{3/2}$ from $x = 0$ to $x = 3$.
 - 2 Set up (do not evaluate) the arc length integral for $y = \sin x$ on $[0, \pi]$.
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Improper integrals

- 3 Evaluate $\int_1^{\infty} \frac{1}{x^2} dx$, or state that it diverges.
 - 4 Evaluate $\int_1^{\infty} \frac{1}{x} dx$, or state that it diverges.
 - 5 Evaluate $\int_0^1 \frac{1}{\sqrt{x}} dx$ (improper at the lower limit).
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More arc length practice

- 6 Find the arc length of the line $y = 3x + 1$ from $x = 0$ to $x = 4$, and verify your answer using the distance formula.
 - 7 Set up (do not evaluate) the arc length integral for $y = \ln x$ on $[1, e]$.
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More improper integrals

- 8 Evaluate $\int_1^{\infty} \frac{1}{x^{1.5}} dx$, or state that it diverges.
 - 9 Evaluate $\int_{-\infty}^0 e^x dx$, or state that it diverges.
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The p-integral test

- 10 For $\int_1^{\infty} \frac{1}{x^p} dx$, state the condition on p for convergence, and explain why using the antiderivative $\frac{x^{1-p}}{1-p}$ (for $p \neq 1$).
 - 11 Use the p -integral result to quickly determine whether $\int_1^{\infty} \frac{1}{x^{0.9}} dx$ converges.
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Improper integrals with infinite discontinuities inside the interval

- 12** Evaluate $\int_0^2 \frac{1}{(x-1)^2} dx$, or explain why it diverges (note the integrand is undefined at $x = 1$, inside the interval).
- 13** Evaluate $\int_{-1}^1 \frac{1}{x^{1/3}} dx$ (an odd function with a discontinuity at $x = 0$), splitting at $x = 0$.